

```
Clear ["Global`*"]
```

```
m1 = {{1, d1}, {0, 1}} (*1st free space matrix*)
```

```
{{1, d1}, {0, 1}}
```

```
m2 = {{1, 0}, {-1/f1, 1}} (*1st lens matrix*)
```

```
{{1, 0}, {-1/f1, 1}}
```

```
m3 = {{1, d2}, {0, 1}} (*2nd free space matrix*)
```

```
{{1, d2}, {0, 1}}
```

```
m4 = {{1, 0}, {-1/f2, 1}} (*2nd lens matrix*)
```

```
{{1, 0}, {-1/f2, 1}}
```

```
m5 = {{1, d3}, {0, 1}} (*3rd free space matrix*)
```

```
{{1, d3}, {0, 1}}
```

```
m6 = {{1, 0}, {-(n2 - n1)/R, 1}} (*thick lens entrance*)
```

```
{{1, 0}, {(n1 - n2)/R, 1}}
```

```
m7 = {{1, d4/n2}, {0, 1}} (*4th free space matrix - through thick lens*)
```

```
{{1, d4/n2}, {0, 1}}
```

```
m8 = {{1, 0}, {(n1 - n2)/R, 1}} (*thick lens exit*)
```

```
{{1, 0}, {(n1 - n2)/R, 1}}
```

```
m9 = {{1, d5}, {0, 1}} (*5th free space matrix*)
```

```
{{1, d5}, {0, 1}}
```

```
m10 = {{1, 0}, {-1/f3, 1}} (*3rd lens matrix*)
```

```
{{1, 0}, {-1/f3, 1}}
```

```
m11 = {{1, d6}, {0, 1}} (*6th free space matrix*)
```

```
{{1, d6}, {0, 1}}
```

```
result = m11.m10.m9.m8.m7.m6.m5.m4.m3.m2.m1
```

$$\left\{ \left\{ 1 - \frac{d6}{f3} - \frac{1}{f2} \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{d4 \left(1 - \frac{d6}{f3} + \frac{(d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2)}{R} \right)}{n2} + d3 \left(1 - \frac{d6}{f3} + \frac{(d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2)}{R} \right) + \frac{1}{R} \right. \right. \right.$$

$$\begin{aligned}
& \left. \left. \left. (n1 - n2) \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{1}{n2} d4 \left(1 - \frac{d6}{f3} + \frac{1}{R} \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) \right) (n1 - n2) \right) \right) \right) \right) \right) - \\
& \frac{1}{f1} \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{d4 \left(1 - \frac{d6}{f3} + \frac{(d6+d5 \left(1 - \frac{d6}{f3} \right)) (n1-n2)}{R} \right)}{n2} + \right. \\
& d3 \left(1 - \frac{d6}{f3} + \frac{(d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2)}{R} + \frac{1}{R} \right. \\
& \left. \left. \left. (n1 - n2) \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{1}{n2} d4 \left(1 - \frac{d6}{f3} + \frac{1}{R} \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) \right) (n1 - n2) \right) \right) \right) \right) \right) + \\
& d2 \left(1 - \frac{d6}{f3} - \frac{1}{f2} \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{1}{n2} d4 \left(1 - \frac{d6}{f3} + \frac{1}{R} \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) \right) (n1 - n2) \right) + \right. \right. \\
& d3 \left(1 - \frac{d6}{f3} + \frac{1}{R} \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) \right) (n1 - n2) + \frac{1}{R} (n1 - n2) \right. \\
& \left. \left. \left. \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{1}{n2} d4 \left(1 - \frac{d6}{f3} + \frac{1}{R} \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) \right) (n1 - n2) \right) \right) \right) \right) \right) + \\
& \left. \frac{(d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{1}{n2} \right. \right. \\
& \left. \left. d4 \left(1 - \frac{d6}{f3} + \frac{1}{R} \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) \right) (n1 - n2) \right) \right) \right) \right) + \\
& \left. \frac{(d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \right. \right. \\
& \left. \left. \frac{d4 \left(1 - \frac{d6}{f3} + \frac{(d6+d5 \left(1 - \frac{d6}{f3} \right)) (n1-n2)}{R} \right)}{n2} \right) \right), \\
& d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{d4 \left(1 - \frac{d6}{f3} + \frac{(d6+d5 \left(1 - \frac{d6}{f3} \right)) (n1-n2)}{R} \right)}{n2} + \\
& d3 \left(1 - \frac{d6}{f3} + \frac{(d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2)}{R} + \frac{1}{R} \right. \\
& \left. \left. \left. (n1 - n2) \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{d4 \left(1 - \frac{d6}{f3} + \frac{(d6+d5 \left(1 - \frac{d6}{f3} \right)) (n1-n2)}{R} \right)}{n2} \right) \right) \right) \right) +
\end{aligned}$$

$$\begin{aligned}
& d2 \left(1 - \frac{d6}{f3} - \frac{1}{f2} \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{d4 \left(1 - \frac{d6}{f3} + \frac{(d6+d5 \left(1 - \frac{d6}{f3} \right)) (n1-n2)}{R} \right)}{n2} \right) + \right. \\
& \quad d3 \left(1 - \frac{d6}{f3} + \frac{(d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{1}{n2} d4 \left(1 - \right. \right. \\
& \quad \left. \left. \frac{d6}{f3} + \frac{1}{R} (d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2) \right) \right) \right) + \frac{(d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2)}{R} + \\
& \quad \left. \frac{1}{R} (n1 - n2) \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{d4 \left(1 - \frac{d6}{f3} + \frac{(d6+d5 \left(1 - \frac{d6}{f3} \right)) (n1-n2)}{R} \right)}{n2} \right) \right) + \\
& \quad d1 \left(1 - \frac{d6}{f3} - \frac{1}{f2} \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{d4 \left(1 - \frac{d6}{f3} + \frac{(d6+d5 \left(1 - \frac{d6}{f3} \right)) (n1-n2)}{R} \right)}{n2} \right) + \right. \\
& \quad d3 \left(1 - \frac{d6}{f3} + \frac{(d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \right. \\
& \quad \left. \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{1}{n2} d4 \left(1 - \frac{d6}{f3} + \frac{1}{R} (d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2) \right) \right) \right) \right) - \\
& \quad \frac{1}{f1} \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{d4 \left(1 - \frac{d6}{f3} + \frac{(d6+d5 \left(1 - \frac{d6}{f3} \right)) (n1-n2)}{R} \right)}{n2} + \right. \\
& \quad d3 \left(1 - \frac{d6}{f3} + \frac{(d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \right. \\
& \quad \left. \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{1}{n2} d4 \left(1 - \frac{d6}{f3} + \frac{1}{R} (d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2) \right) \right) \right) + d2 \\
& \quad \left(1 - \frac{d6}{f3} - \frac{1}{f2} \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{1}{n2} d4 \left(1 - \frac{d6}{f3} + \frac{1}{R} (d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2) \right) \right) + \right. \\
& \quad d3 \left(1 - \frac{d6}{f3} + \frac{1}{R} (d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2) + \frac{1}{R} (n1 - n2) \left(d6 + \right. \\
& \quad \left. d5 \left(1 - \frac{d6}{f3} \right) + \frac{1}{n2} d4 \left(1 - \frac{d6}{f3} + \frac{1}{R} (d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2) \right) \right) \right) + \\
& \quad \left. \frac{(d6 + d5 \left(1 - \frac{d6}{f3} \right)) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{1}{n2} d4 \left(1 - \frac{d6}{f3} + \right. \right. \right.
\end{aligned}$$

$$\begin{aligned}
& \left. \left. \left. \frac{1}{R} \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) \right) (n1 - n2) \right) \right) \right) + \frac{\left(d6 + d5 \left(1 - \frac{d6}{f3} \right) \right) (n1 - n2)}{R} + \\
& \frac{1}{R} (n1 - n2) \left(d6 + d5 \left(1 - \frac{d6}{f3} \right) + \frac{d4 \left(1 - \frac{d6}{f3} + \frac{(d6+d5 \left(1 - \frac{d6}{f3} \right)) (n1-n2)}{R} \right)}{n2} \right) \right) \Bigg\}, \\
& \left\{ -\frac{1}{f3} - \frac{1}{f2} \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{(1-\frac{d5}{f3}) (n1-n2)}{R} \right)}{n2} \right) + \right. \\
& d3 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3} \right) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{(1-\frac{d5}{f3}) (n1-n2)}{R} \right)}{n2} \right) \right) \Bigg\} - \\
& \frac{1}{f1} \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{(1-\frac{d5}{f3}) (n1-n2)}{R} \right)}{n2} + \right. \\
& d3 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3} \right) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{(1-\frac{d5}{f3}) (n1-n2)}{R} \right)}{n2} \right) \right) \Bigg\} + \\
& d2 \left(-\frac{1}{f3} - \frac{1}{f2} \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{(1-\frac{d5}{f3}) (n1-n2)}{R} \right)}{n2} \right) + d3 \right. \\
& \left. \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3} \right) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{(1-\frac{d5}{f3}) (n1-n2)}{R} \right)}{n2} \right) \right) \right) \Bigg\} + \\
& \frac{\left(1 - \frac{d5}{f3} \right) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{(1-\frac{d5}{f3}) (n1-n2)}{R} \right)}{n2} \right) \Bigg\} + \\
& \frac{\left(1 - \frac{d5}{f3} \right) (n1 - n2)}{R} + \frac{(n1 - n2) \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{(1-\frac{d5}{f3}) (n1-n2)}{R} \right)}{n2} \right)}{R}, \\
& 1 - \\
& \frac{d5}{f3} + \\
& \frac{d4 \left(-\frac{1}{f3} + \frac{(1-\frac{d5}{f3}) (n1-n2)}{R} \right)}{n2} + \\
& d3
\end{aligned}$$

$$\begin{aligned}
& \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} \right)}{n2} \right) \right) + \\
& d2 \left(-\frac{1}{f3} - \frac{1}{f2} \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} \right)}{n2} \right) + \right. \\
& \quad \left. d3 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} \right)}{n2} \right) \right) \right) + \\
& \quad \left. \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} \right)}{n2} \right) \right) + \\
& d1 \left(-\frac{1}{f3} - \frac{1}{f2} \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} \right)}{n2} \right) + \right. \\
& \quad \left. d3 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} \right)}{n2} \right) \right) \right) - \\
& \quad \frac{1}{f1} \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} \right)}{n2} \right) + \\
& \quad d3 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} \right)}{n2} \right) \right) + \\
& \quad d2 \left(-\frac{1}{f3} - \frac{1}{f2} \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} \right)}{n2} \right) + d3 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} + \right. \right. \\
& \quad \left. \left. \frac{1}{R} (n1 - n2) \left(1 - \frac{d5}{f3} + \frac{1}{n2} d4 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} \right) \right) \right) \right) + \\
& \quad \left. \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} \right)}{n2} \right) \right) \right) +
\end{aligned}$$

$$\frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R} + \frac{1}{R} (n1 - n2) \left(1 - \frac{d5}{f3} + \frac{d4 \left(-\frac{1}{f3} + \frac{\left(1 - \frac{d5}{f3}\right) (n1 - n2)}{R}\right)}{n2}\right) \Bigg) \Bigg\} \Bigg\}$$

d1 = 0.03

f1 = 0.08

d2 = 0.05

f2 = 0.06

d3 = 0.03

n1 = 1

n2 = 1.6

R = 0.024

d4 = 0.08

d5 = 0.05

f3 = 0.05

d6 = 0.02

0.03

0.08

0.05

0.06

0.03

1

1.6

0.024

0.08

0.05

0.05

0.02

result

{{-0.124219, -0.0606224}, {17.8125, 0.642708}}